Self-Check

Lecture 0

Prerequisites: A Refresher

This sheet begins with a short diagnostic of the four computational skills the early lectures rely on. Take only the seven problems listed below *cold*, without notes, and stop after 45 minutes. Then check your work and use the routing guide at the end: repair only the weak strand and complete its recheck. The remaining algebra problems form a targeted bank, not a second cold test. Throughout, $\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$.

Scored algebra diagnostic (40–45 minutes). Attempt problems **0.1, 0.2, 0.9, 0.15, 0.22, 0.23, and 0.30**. Award one point only for a correct result supported by enough working to identify the method. These seven checks cover complex arithmetic and polar form, compatible matrix operations, determinants, row reduction, null-space bases, and consistency-first classification.

- **Ready:** at least 6/7, including 0.9, 0.15, and 0.30, plus at least one correct complex check (0.1–0.2) and one correct row-reduction check (0.22–0.23).
- **Targeted repair:** 4–5 points, or any missing required strand above. Use the matching refresher section and bank route, then complete the named recheck before Lecture 1.
- **Rebuild first:** 0–3 points, or gaps in two or more strands. Work through each weak route with notes open, then retake the seven diagnostic checks on a later day.

Separate analysis-readiness route. Problems 0.32–0.36 are a non-scored, 15–20 minute routing check for differentiation, integration by parts, elementary linear ODEs, series, and Fourier coefficients. They are not part of the seven-point algebra score and are not assumed in Lecture 0 itself. Use them before the later application weeks; this sheet is not a replacement calculus or analysis course.

Complex arithmetic

Exercise 0.1 (diagnostic) Let $z = 3 + 2i$ and $w = 1 + i$. Compute $z + w$, zw, z/w, the conjugate $\overline{z}$, and the modulus $|z|$.

Answer. $z + w = 4 + 3i$; $\quad zw = 1 + 5i$; $\quad z/w = \frac{5}{2} - \frac{1}{2}i$; $\quad \overline{z} = 3 - 2i$; $\quad |z| = \sqrt{13}$.

Exercise 0.2 (diagnostic) Write $1 - i$ in polar form and use it to compute $(1 - i)^8$.

Answer. $1 - i = \sqrt{2}\, e^{-i\pi/4}$, so $(1 - i)^8 = 16$.

Exercise 0.3 (computational) Find all complex numbers z with $z^2 = i$.

Hint. Write i in polar form and take square roots: halve the argument, and remember a quadratic has two roots.

Answer. $z = \pm\dfrac{1 + i}{\sqrt{2}} = \pm\, e^{i\pi/4}$.

Exercise 0.4 (computational) Let $z = 3 + i$ and $w = 1 + 2i$. Compute $z + w$, zw, z/w, the conjugate $\overline{z}$, and the modulus $|z|$.

Answer. $z + w = 4 + 3i$; $\quad zw = 1 + 7i$; $\quad z/w = 1 - i$; $\quad \overline{z} = 3 - i$; $\quad |z| = \sqrt{10}$.

Exercise 0.5 (computational) Write $1 + i$ in polar form and use it to compute $(1 + i)^6$.

Answer. $1 + i = \sqrt{2}\, e^{i\pi/4}$, so $(1+i)^6 = -8i$.

Exercise 0.6 (computational) Write $\sqrt{3} + i$ in polar form and use it to compute $(\sqrt{3} + i)^4$, giving the answer in Cartesian form.
Answer. $\sqrt{3} + i = 2\, e^{i\pi/6}$, so $(\sqrt{3} + i)^4 = -8 + 8\sqrt{3}\, i$.

Exercise 0.7 (bridge) For complex numbers $z = a + bi$ and $w = c + di$, prove that $\overline{zw} = \overline{z}\,\overline{w}$.

Exercise 0.8 (extension) Find all four complex numbers z with $z^4 = -4$.
Hint. Write -4 in polar form as $4\, e^{i\pi}$; a fourth root has modulus $4^{1/4} = \sqrt{2}$ and argument $\frac{1}{4}(\pi + 2\pi k)$ for $k = 0, 1, 2, 3$.

Answer. $z \in \{1+i,\ -1+i,\ -1-i,\ 1-i\}$, i.e. $z = \pm 1 \pm i$.

Matrix algebra

Exercise 0.9 (diagnostic) Let $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$. Compute AB, BA, and A^{T}, and verify $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ even though $AB \neq BA$.
Answer. $AB = \begin{pmatrix} 5 & 5 \\ -1 & -2 \end{pmatrix}$, $BA = \begin{pmatrix} 3 & 5 \\ 1 & 0 \end{pmatrix}$, $A^{\mathsf{T}} = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix}$; $\operatorname{tr}(AB) = 3 = \operatorname{tr}(BA)$.

Exercise 0.10 (computational) Let $A = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$. Compute AB and BA, and verify $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ although $AB \neq BA$.
Answer. $AB = \begin{pmatrix} 2 & -2 \\ 7 & 2 \end{pmatrix}$, $BA = \begin{pmatrix} 1 & -3 \\ 5 & 3 \end{pmatrix}$; $\operatorname{tr}(AB) = 4 = \operatorname{tr}(BA)$.

Exercise 0.11 (computational) For $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, compute A^2 and A^3.

Answer. $A^2 = \begin{pmatrix} 4 & 4 \\ 0 & 4 \end{pmatrix}$, $A^3 = \begin{pmatrix} 8 & 12 \\ 0 & 8 \end{pmatrix}$.

Exercise 0.12 (bridge) Let $A = \begin{pmatrix} 1+i & 2-i \\ 0 & 3i \end{pmatrix}$ over $\mathbb{C}$. Write down the transpose A^{T} and the conjugate transpose $A^{\dagger} = \overline{A}^{\mathsf{T}}$.
Answer. $A^{\mathsf{T}} = \begin{pmatrix} 1+i & 0 \\ 2-i & 3i \end{pmatrix}$, $A^{\dagger} = \begin{pmatrix} 1-i & 0 \\ 2+i & -3i \end{pmatrix}$.

Exercise 0.13 (extension) For square matrices A, B of the same size, prove the cyclic identity $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Exercise 0.14 (bridge) For matrices $A \in \mathcal{M}_{m\times n}(\mathbb{F})$ and $B \in \mathcal{M}_{n\times p}(\mathbb{F})$, prove that $(AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}$.

Determinants

Exercise 0.15 (diagnostic) Evaluate $\det\begin{pmatrix}1 & 2 & 3\\ 0 & 1 & 4\\ 5 & 6 & 0\end{pmatrix}$ and state whether the matrix is invertible.

Answer. $\det = 1$; invertible.

Exercise 0.16 (computational) Use the 2×2 formula to invert $A = \begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$.

Answer. $A^{-1} = \begin{pmatrix}-2 & 1\\ \frac{3}{2} & -\frac{1}{2}\end{pmatrix}$.

Exercise 0.17 (bridge) For $A = \begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}$, find all λ for which $\det(\lambda I - A) = 0$. (These values reappear in Week 5 as the eigenvalues of A.)

Answer. $\lambda = 1$ and $\lambda = 3$.

Exercise 0.18 (computational) Evaluate $\det\begin{pmatrix}1 & 2 & 1\\ 0 & 3 & 2\\ 1 & 0 & 1\end{pmatrix}$ and state whether the matrix is invertible.

Answer. $\det = 4$; invertible.

Exercise 0.19 (computational) Let $A = \begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$ and $B = \begin{pmatrix}2 & 0\\ 1 & 3\end{pmatrix}$. Compute $\det A$, $\det B$, and $\det(AB)$, and confirm $\det(AB) = \det A\,\det B$.

Answer. $\det A = -2$, $\det B = 6$, $\det(AB) = -12 = (-2)(6)$.

Exercise 0.20 (extension) Evaluate the block-diagonal determinant $\det\begin{pmatrix}1 & 1 & 0 & 0\\ 1 & 2 & 0 & 0\\ 0 & 0 & 2 & 0\\ 0 & 0 & 1 & 3\end{pmatrix}$.

Hint. A determinant with a zero off-diagonal block factors as the product of the determinants of the diagonal blocks.

Answer. $\det = 6$.

Exercise 0.21 (extension) For 2×2 matrices A, B, prove that $\det(AB) = \det A\,\det B$ directly from the entries.

Gaussian elimination

Exercise 0.22 (diagnostic) Solve, by row reduction,

$$\begin{aligned} x + y + z &= 2,\\ x - y + 2z &= 5,\\ 2x + y - z &= 2. \end{aligned}$$

Answer. $(x, y, z) = (2, -1, 1)$ (unique).

Exercise 0.23 (diagnostic) Find a basis for the solution space (null space) of $Ax = 0$, where $A = \begin{pmatrix} 1 & -1 & 2 & 0 \\ 2 & -2 & 3 & 1 \end{pmatrix}$, and state its dimension.

Hint. Row-reduce A, identify the pivot and free columns, then solve for the pivot variables in terms of the free ones.

Answer. Basis $\{(1,1,0,0),\ (-2,0,1,1)\}$; dimension 2.

Exercise 0.24 (computational) Compute A^{-1} by row-reducing the augmented block $[A \mid I]$, for $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$.

Answer. $A^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$.

Exercise 0.25 (computational) Solve, by row reduction,

$$\begin{aligned} x + y + z &= 6, \\ x - y + z &= 2, \\ 2x + y - z &= 1. \end{aligned}$$

Answer. $(x, y, z) = (1, 2, 3)$ (unique).

Exercise 0.26 (computational) Describe the full solution set of

$$\begin{aligned} x + 2y - z &= 3, \\ 2x + y + z &= 3, \end{aligned}$$

in the form "particular solution + span of the null space".

Answer. $(x, y, z) = (1, 1, 0) + t(-1, 1, 1)$, $t \in \mathbb{R}$.

Exercise 0.27 (computational) Compute A^{-1} by row-reducing the augmented block $[A \mid I]$, for $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

Answer. $A^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$.

Exercise 0.28 (bridge) Let $A \in \mathcal{M}_{m\times n}(\mathbb{F})$. Prove that the null space $\operatorname{null} A = \{x \in \mathbb{F}^n : Ax = 0\}$ is closed under addition and scalar multiplication (so it is a subspace of $\mathbb{F}^n$).

Exercise 0.29 (extension) For which real values of a does the system

$$\begin{aligned} ax + y + z &= 1, \\ x + ay + z &= 1, \\ x + y + az &= 1 \end{aligned}$$

have a unique solution, infinitely many solutions, or no solution? Give the solution in the unique case.

Hint. Add the three equations to get $(a+2)(x+y+z) = 3$; subtract pairs to get $(a-1)(x-y) = 0$ and $(a-1)(y-z) = 0$. The determinant of the coefficient matrix is $(a-1)^2(a+2)$.

Answer. Unique iff $a \neq 1$ and $a \neq -2$, with $x = y = z = \dfrac{1}{a+2}$; infinitely many if $a = 1$ (the plane $x + y + z = 1$); no solution if $a = -2$.

Exercise 0.30 (diagnostic) Classify each displayed RREF augmented matrix as having no solution, a unique solution, or infinitely many solutions. For each consistent system, give its full solution set.

$$\text{(a)} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array}\right], \qquad \text{(b)} \left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 1 \end{array}\right], \qquad \text{(c)} \left[\begin{array}{ccc|c} 1 & 0 & -2 & 4 \\ 0 & 1 & 3 & -1 \end{array}\right].$$

Answer. (a) Unique: $(2, -1)$. (b) No solution. (c) Infinitely many: $(x, y, z) = (4, -1, 0) + t(2, -3, 1)$, $t \in \mathbb{R}$.

Exercise 0.31 (repair) Row-reduce and classify each system, giving the full solution set when it is consistent:

$$\begin{array}{lll} \text{(a)} & x + y = 1, & x - y = 3, \\ \text{(b)} & x + y = 1, & 2x + 2y = 3, \\ \text{(c)} & x + y = 1, & 2x + 2y = 2. \end{array}$$

Answer. (a) Unique: $(2, -1)$. (b) No solution. (c) Infinitely many: $(x, y) = (1, 0) + t(-1, 1)$, $t \in \mathbb{R}$.

Separate analysis-readiness route

Take these five checks without notes in 15–20 minutes before the application weeks. They do not contribute to the algebra-readiness score.

Exercise 0.32 (analysis route) Differentiate $f(t) = t^2 e^{-t}$.

Answer. $f'(t) = e^{-t}(2t - t^2)$.

Exercise 0.33 (analysis route) Use integration by parts to evaluate $\displaystyle\int_0^1 te^t\,dt$.

Answer. 1.

Exercise 0.34 (analysis route) Find the general real solution of $y'' - 3y' + 2y = 0$.

Answer. $y(t) = C_1 e^t + C_2 e^{2t}$.

Exercise 0.35 (analysis route) Determine whether $\displaystyle\sum_{n=0}^{\infty} 3^{-n}$ converges and, if it does, find its sum.

Answer. It converges to $\dfrac{3}{2}$.

Exercise 0.36 (analysis route) For $f(x) = x$ on $[-\pi, \pi]$, compute the first Fourier sine coefficient $b_1 = \dfrac{1}{\pi}\displaystyle\int_{-\pi}^{\pi} x \sin x\,dx$.

Answer. $b_1 = 2$.

Reading your results

Follow only the route for a missed diagnostic strand. Work with the refresher open, close it for the named recheck, and require a correct result with working before declaring the strand repaired.

- **Complex arithmetic** (diagnostic 0.1–0.2): revisit *Refresher* §1; practise 0.3–0.6; recheck with 0.4 and 0.5. Problem 0.7 is a proof bridge and 0.8 is extension, neither scored for readiness.
- **Matrix algebra** (diagnostic 0.9): revisit *Refresher* §2; practise and recheck with 0.10–0.11. Problem 0.12 is an adjoint bridge and 0.13–0.14 are proof extensions/bridges, excluded from readiness.
- **Determinants** (diagnostic 0.15): revisit *Refresher* §3; practise 0.16 and 0.18–0.19; recheck with 0.18. Problem 0.17 is the Week 5 characteristic-polynomial bridge; 0.20–0.21 are extensions.
- **Gaussian elimination** (diagnostic 0.22–0.23 and 0.30): revisit *Refresher* §4; practise 0.24–0.27; recheck consistency and full solution sets with 0.31. Problem 0.28 is a Lecture 1 subspace bridge and 0.29 is a parameter-family extension, neither scored for readiness.

Analysis routing. For problems 0.32–0.36, 4–5 correct means the later application bridge is ready; 2–3 means repair only the named topics; and 0–1 means arrange a calculus/methods review before those weeks. Route 0.32 to product/chain rules, 0.33 to integration by parts, 0.34 to constant-coefficient ODEs (before Lectures 1 and 6), 0.35 to geometric series (before Lectures 6 and 13), and 0.36 to Fourier coefficients (before Lecture 13). Use prior calculus/methods notes or tutorial support, work two similar examples, then redo the missed check without notes on a later day.